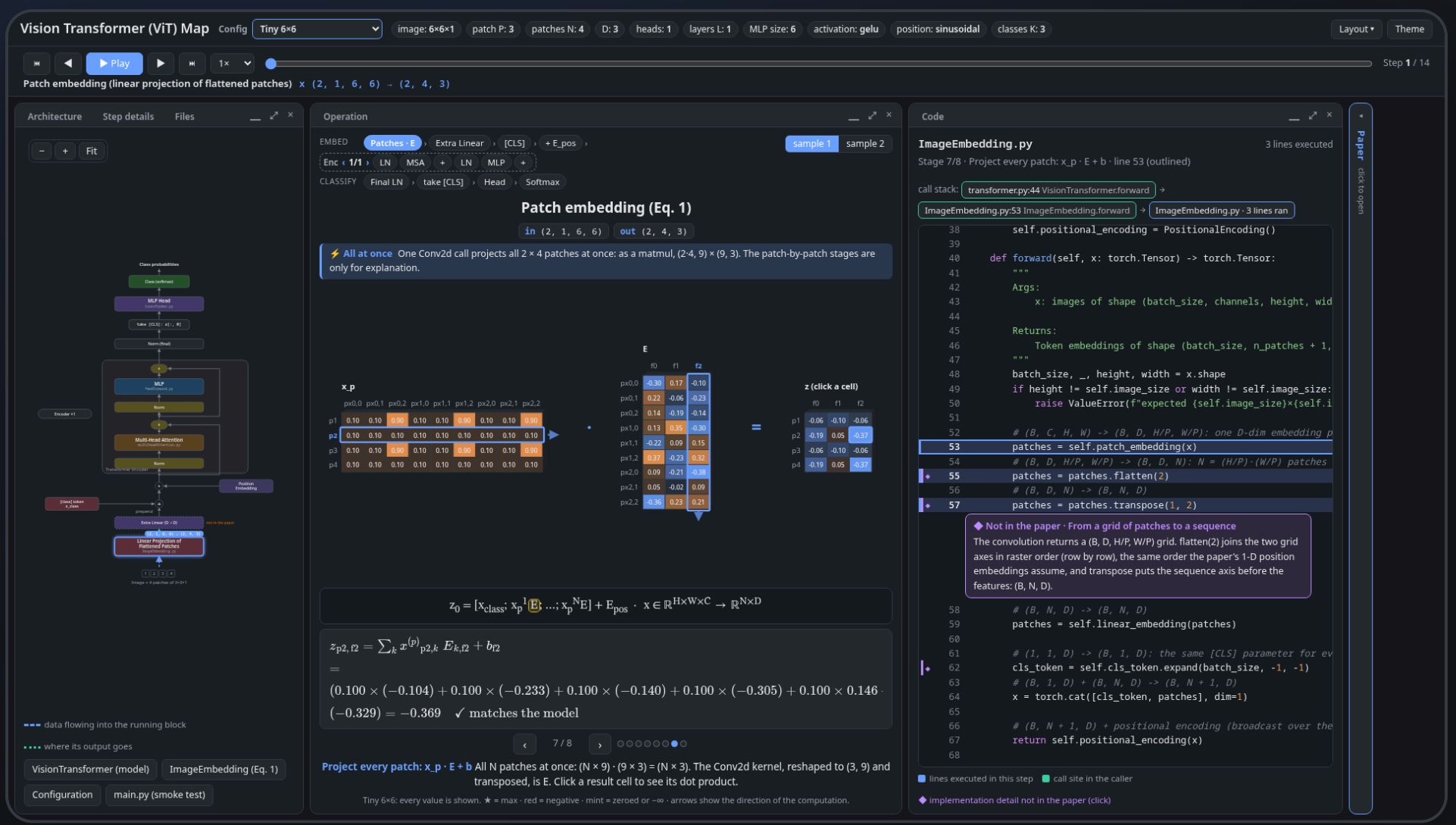
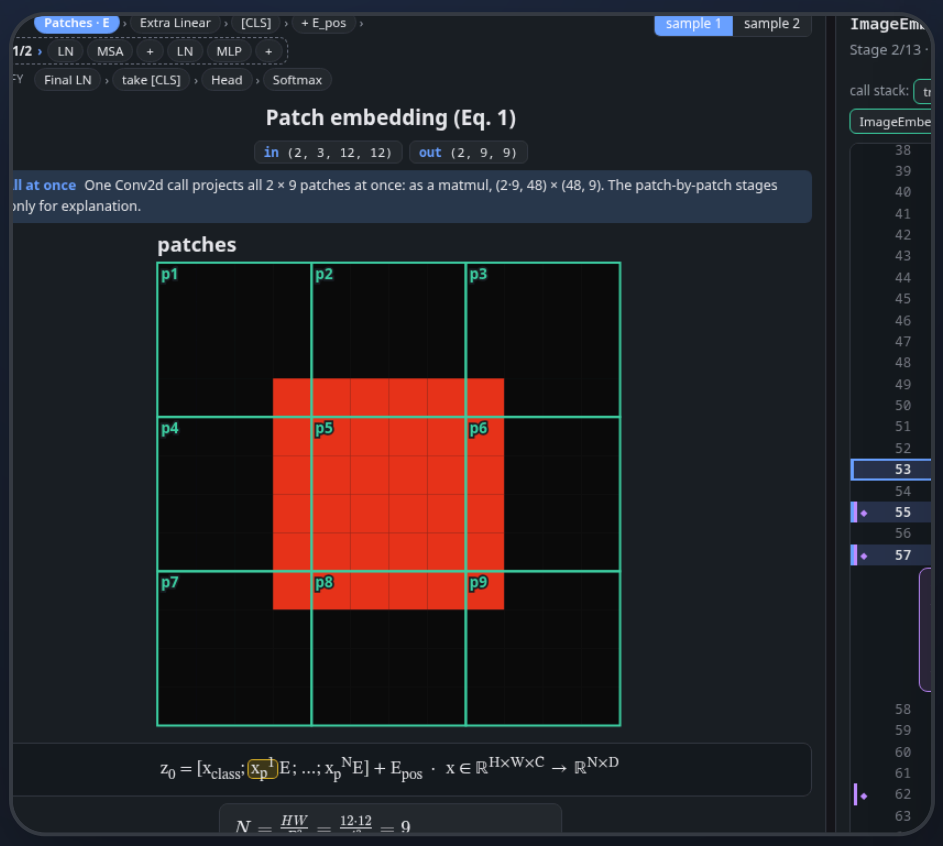
An image is worth 16×16 words

An interactive walkthrough of a real PyTorch Vision Transformer: pixels → patches → tokens → class, one tensor operation at a time, linked to the exact line of code that runs it.



Swipe →

Every step, worked out



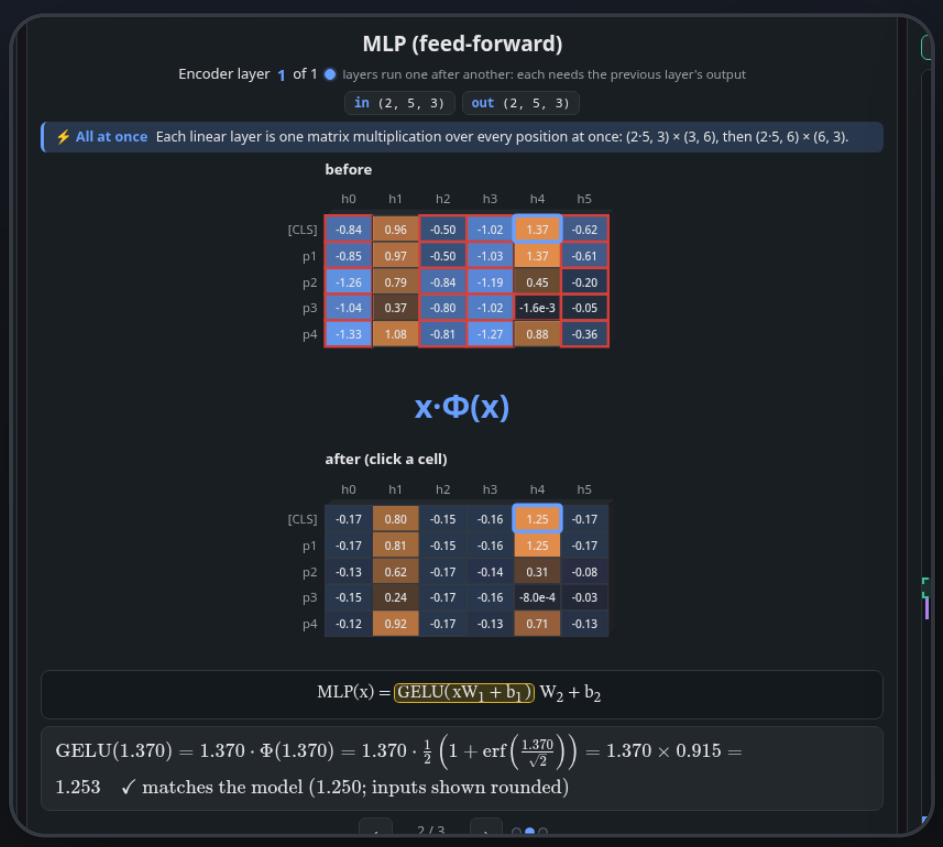
Pixels to patches

The image cut into P×P patches in raster order, each flattened to P²·C values.



Conv2d = x_p·E

The patch projection as a matmul: click any cell to see its dot product, checked against the model.



LaTeX arithmetic

GELU with erf, LayerNorm, softmax and the residual adds, written out with the real values.

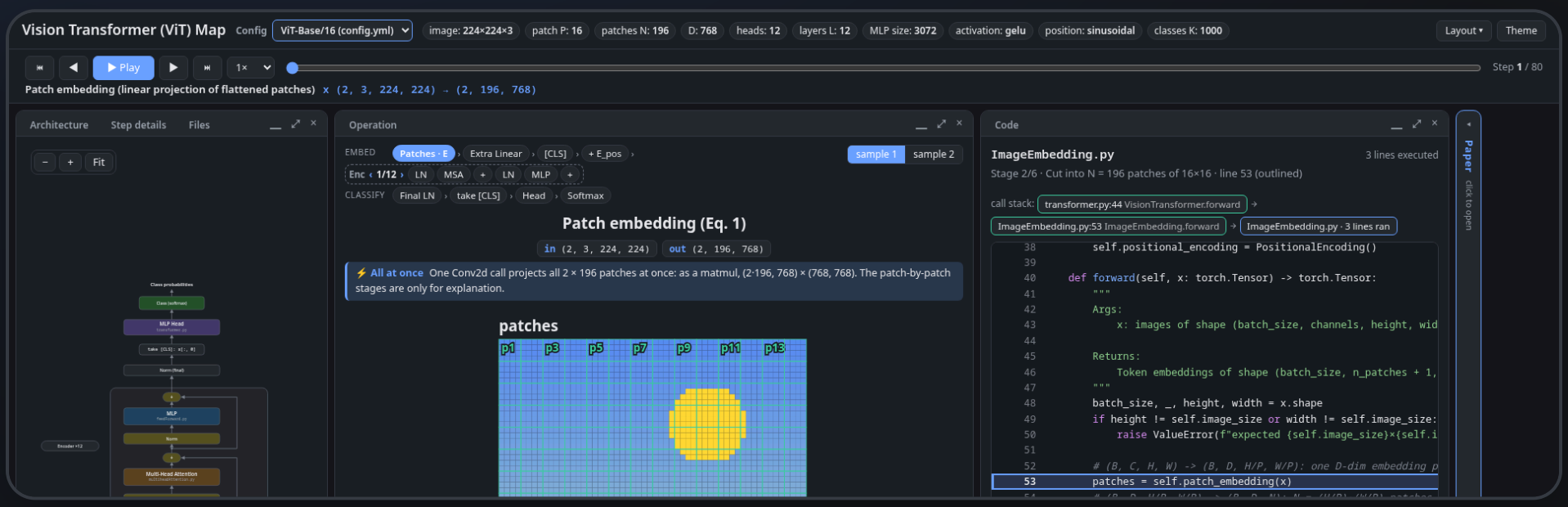


Honest about the code

What this implementation adds beyond the paper (an extra Linear, sinusoidal positions) is flagged.

Small enough to check by hand

- **Three sizes:** Tiny 6×6 (4 patches, D = 3: every value on screen), Small 12×12 RGB (3 heads, 2 layers) and ViT-Base/16.
- **Real execution trace:** captured from the PyTorch model with hooks and sys.settrace; nothing is hand-drawn.
- **Pre-norm encoder:** LN → MSA → + → LN → MLP → +, then [CLS] → head → softmax, exactly as in Eq. 1–4.
- **Same workspace as the encoder–decoder:** dockable panels, architecture diagram, code and paper notes in sync.



`github.com/omarTBakr/TransformersVisualizations`

Open source. Also: the encoder–decoder Transformer. Next: decoder-only (GPT-style).